Project: Worked examples
Title: Fatigue damage (Palmgren–Miner)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Cumulative fatigue damage of a stress-range spectrum against an S–N curve of one slope through a reference point, with an optional endurance limit below which a range does no damage; the spectrum repetitions to failure.

WHAT IT CALCULATES	RELATION
Cycles to failure at each stress range, from the S–N curve (unbounded at or below the endurance limit S_e)	$N_i = N_{ref} \cdot (S_{ref}/S_i)^m$
Damage done by each block of cycles	$d_i = n_i/N_i$
Total damage, and how many repetitions of the spectrum the part survives	$D = \Sigma\, d_i$; repetitions = 1/D
Acceptance by the Palmgren–Miner rule	$D \le 1$

Basis: Fatigue resistance (S–N curves by detail category)[1](§8); Cumulative fatigue damage[2](§6-15).

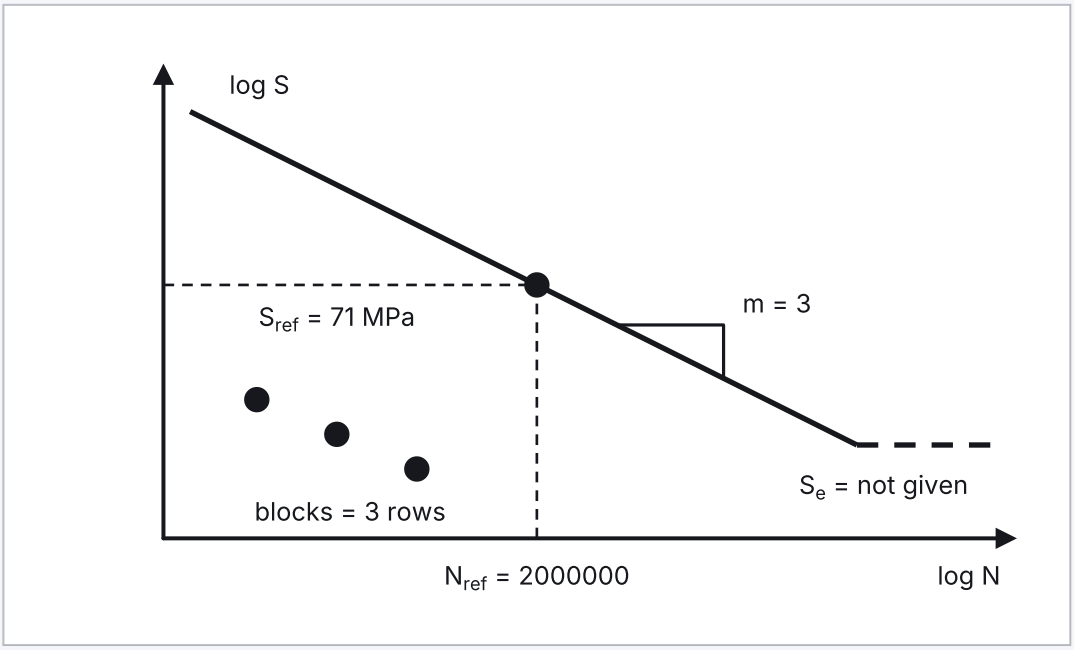


Figure 1 — A single-slope S–N line on log axes through its reference point, the endurance limit as a cut-off, and the stress-range blocks as listed (three drawn, representatively). Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
S_{ref}	Reference stress range (detail category or material fatigue strength)	71	MPa	spec: SPEC-MECH-03 design criteria	1 … 2000 MPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
N_{ref}	Cycles at the reference stress range	2000000		spec: SPEC-MECH-03 design criteria	10000 … 1000000000
m	S–N slope	3		spec: SPEC-MECH-03 design criteria	2 … 15
blocks	Stress-range blocks	3 row(s)		spec: LOAD-SCHEDULE-01	any

TABLE ROW	VALUES
blocks[1]	S = 100 MPa; n = 100000
blocks[2]	S = 60 MPa; n = 500000
blocks[3]	S = 40 MPa; n = 2000000

3. Assumptions

A1. Damage accumulates linearly and independently of the order of the blocks. (Palmgren–Miner rule.)
A2. One S–N slope over the whole range; no second slope below a knee. (Simplified curve; a two-slope curve is entered as two sheets or a lower slope.)

4. Working

[4] $N_1 = N_{ref} \cdot (S_{ref} / S_1)^m$ [1](§8)
= 2000000 × (71 MPa / 100 MPa)^3
= 715820
Life at block 1

[5] $d_1 = n_1 / N_1$
= 100000 / 715820
= 0.1397
Damage of block 1

[6] $N_2 = N_{ref} \cdot (S_{ref} / S_2)^m$ [1](§8)
= 2000000 × (71 MPa / 60 MPa)^3
= 3314000
Life at block 2

[7] $d_2 = n_2 / N_2$
= 500000 / 3314000
= 0.15088
Damage of block 2

[8] $N_3 = N_{ref} \cdot (S_{ref} / S_3)^m$ [1](§8)
= 2000000 × (71 MPa / 40 MPa)^3
= 11185000
Life at block 3

[9] $d_3 = n_3 / N_3$
= 2000000 / 11185000
= 0.17882
Damage of block 3

[10] $D = \Sigma\, d_i$ [2](§6-15)
= 0.1397 + 0.15088 + 0.17882
= 0.46939
Total damage (Palmgren–Miner)

[11] $R_f = 1 / D$
$= 1 / 0.46939$
$= 2.1304$
Spectrum repetitions to failure

5. Results

SYMBOL	VALUE	DESCRIPTION
R_f	2.1304	Repetitions to failure
N_1	715820	Life at block 1
d_1	0.1397	Damage of block 1
N_2	3314000	Life at block 2
d_2	0.15088	Damage of block 2
N_3	11185000	Life at block 3
d_3	0.17882	Damage of block 3
D	0.47	Total damage (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Cumulative damage D ≤ 1	D = 0.47	D ≤ 1	0.47	[2](§6-15)

7. Applicability

None.

8. Verification

Template mech.fatigue-miner@1.0.1 Content hash 82e9e2cff5d85046 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-9: Fatigue. BS EN 1993-1-9. BSI.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.